Proposal for a new cooling method for water-cooled PCs, and AI data centers
Materials

Obtaining a confirmed date 確定日付取得
2026.04.01Wednesday About am04:30

Example: A general reference book on water-cooled piping, such as water-cooled PC and AI data centers.

Summary 1 (1/2)

1. Purpose of this research experiment
This is an ongoing observational experiment aimed at discovering the optimal hose pipe diameter (inner diameter) for small waterways (e.g., server cooling in AI data centers).
Observation period: Approximately 4 days, and repeated observations are made at this interval.
Pump power OFF: Completely turned off.
Event confirmation: Airlock: We have succeeded in quantitatively confirming that an air gap of approximately 15mm to 25mm per day occurs (attached photo).

2. General considerations (causes occurring)
This phenomenon is known as the process by which tiny, invisible bubbles (such as microbubbles) in water collect at specific locations and coalesce into larger bubbles.
Carbon dioxide (carbon dioxide) is one of the main causes because it is easily soluble in water, and dissolved air (nitrogen and oxygen) is said to be the most common cause.
Content rate
The water contains nitrogen (about 78%) and oxygen (about 21%).

3. Relationship to temperature
Rising water temperatures reduce gas solubility. This reduction causes dissolved gases (mainly nitrogen) to precipitate as microbubbles. Eventually, these bubbles coalesce into larger air gaps and create an airlock.

Pump power OFF: The pressure drops in fast-flowing areas and steps due to the effect of reduced pressure, and the dissolved air suddenly returns to gas.
In this way, "nitrogen generated by changes in temperature or pressure" and "spoil gas caused by microorganisms" can be confirmed as visible phenomena known as airlocks.

4. Definitive features of the invention
Capillary＋Master Bag
The airlock described up to said point: It was successfully quantitatively confirmed that an air layer of about 15 mm ～ 25 mm per day was generated. This is an intentional construction of how bubbles form in "Capillary Drain and Retarn" (attached photo).

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ＦＡＸ：０６７６３５２８１５ＴＥＬ：都合によりありません

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Summary 2 (2/2)

5. Predictive effect
By mitigating airlocks, this method enhances the coolant density and ensures complete liquefaction, thereby maximizing heat dissipation efficiency in water-cooling systems.

6. Attached
Timestamped for intellectual property protection.

Materials
・Obtaining a confirmed date 確定日付取得
2026.03.31Tuesday About am10:00

Warning 1
Timestamped for intellectual property protection.
All public information has a confirmed date.
All public information has a confirmed date.

・Photographs of the phenomenon
・Papers explaining the scientific basis.

Warning 2
Timestamped for IP protection
Regarding the date of confirmation of this document.
Regarding the date of confirmation of this document.

Scheduled for acquisition on April 1, 2026 (Wednesday)

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Conclusion 1 (1/2)

Final Technical Conclusion: Optimized Conduit Diameter for Stable Micro-Fluidic Recirculation

1. Abstract
Through extensive R&D and empirical validation, this study identifies the optimal internal conduit diameter for stable gas-liquid phase control in small-scale water channels. The results demonstrate that an internal diameter (ID) of 3 mm to 4 mm provides the most efficient and predictable bubble accumulation and fluid circulation.

2. Physical Basis for Optimization
The superiority of the 3 mm to 4 mm range is governed by the following factors:
Enhanced Energy Capture: At this scale, the balance between flow velocity and negative pressure (Venturi effect) is maximized, facilitating a sharp increase in bubble formation as compared to larger diameters (over 8 mm).
Viscous Force Dominance: In micro-scales (ID under 5 mm), surface tension and viscosity interact with the proprietary mechanical configuration to ensure stable bubble retention, preventing premature coalescence or erratic discharge.

3. Empirical Validation (Standard: 4 mm)
Quantitative analysis confirms that the 4 mm standard maintains a high-efficiency baseline (100% reference), while a 3 mm diameter offers a 115% efficiency gain. This scalability allows for precise system tuning based on specific head (H) and flow rate (Q) requirements.

4. Temporal Stability and Predictability
Long-term testing confirms a highly linear growth rate (approximately 20 to 25 mm/day). The total length of the accumulated air layer (L) is defined by the formula $L = G * t$ (where G is the daily growth rate and t is elapsed time in days). This linearity proves that the mechanism operates at a constant efficiency, unaffected by transient turbulence, which is critical for autonomous, maintenance-free operation.

5. Final Recommendation
For small-scale water systems requiring stable, predictable, and efficient gas-liquid management, the 3 mm to 4 mm range is the technical "sweet spot." This range ensures the highest reliability for engineering applications such as water purification, aeration, and decorative fluidic displays.

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Conclusion 2 (2/2)

Technical Manuscript: Optimization of Small-Scale Water Channels for Stable Gas-Liquid Phase Control

1. Abstract

This paper presents the identification of the optimal internal diameter (ID) for stable bubble accumulation and fluid circulation in small-scale water channels. Through extensive R&D and empirical validation, it is demonstrated that an ID of 3 mm to 4 mm provides the most efficient and predictable performance for specialized gas-liquid mechanisms.

2. Physical Basis and Mechanism

The superiority of the 3 mm–4 mm range is governed by the balance between flow velocity and negative pressure (Venturi effect). In this micro-scale (ID < 5 mm), surface tension and viscosity interact with the proprietary mechanical configuration to ensure stable bubble retention, preventing premature coalescence.

3. Empirical Validation and Efficiency

Quantitative analysis based on experimental data confirms the following efficiency ratios:
3 mm (Minimum): 115% Efficiency (Enhanced energy capture).
4 mm (Standard): 100% Efficiency (Optimal design baseline).
8 mm to 15 mm: 71% to 52% Efficiency (Significant performance decay).

4. Temporal Stability and Linear Growth

Long-term testing confirms a highly stable, linear accumulation of the air layer, reflecting the reliability of the mechanical design:
Day 1: ~15 mm
Day 2 and onwards: Constant growth of 20–25 mm/day.
Mathematical Model:

(where is the daily growth rate and is time).

5. Environmental Factors: Temperature Dependency

The model accounts for water viscosity changes to maintain precision across environments:
5 °C: 0.925 (High viscosity)
20 °C: 1.000 (Standard)
40 °C: 1.100 (Low viscosity)

6. Conclusion

The 3 mm to 4 mm range is the technical "sweet spot" for small-scale water systems requiring autonomous and stable gas-liquid management. This range ensures maximum energy capture and predictable, linear performance, serving as the ideal design standard foradvanced micro-fluidic applications.

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